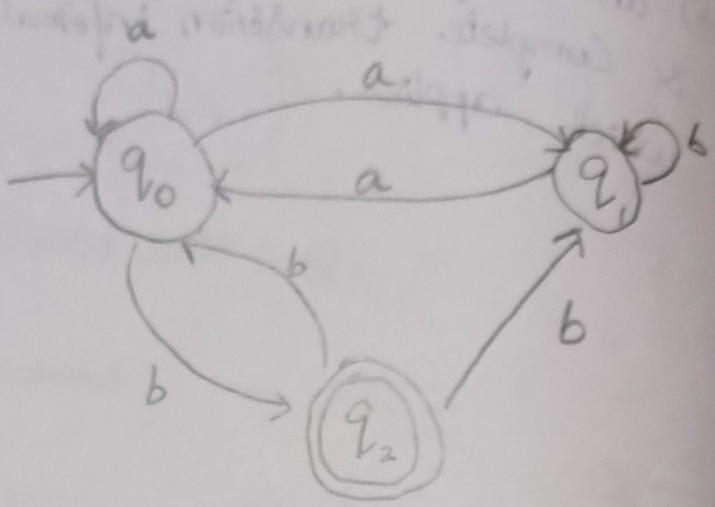
14/10/26

① Consider the NFA given below which accepts some language L over the alphabet Σ. Construct DFA that represents the same language L.



$\Sigma = \{a, b\}$

Initial State : q_0 , Final State : q_2

States : $Q = \{q_0, q_1, q_2\}$

Transitions :

State	Input a	Input b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	ϕ	$\{q_0, q_1\}$

DFA States are set of NFA States

Let, $A = \{q_0\}$ (Initial State)

$B = \{q_0, q_1\}$ $C = \{q_2\}$

$D = \{q_0, q_1, q_2\}$ $E = \{q_1\}$ $F = \phi$

DFA State	Input a	Input b
(Initial state) A	B	C
B	B	D
(Final state) C	F	B
(Final state) D	B	D
E	A	E
F	F	F

Equivalent DFA

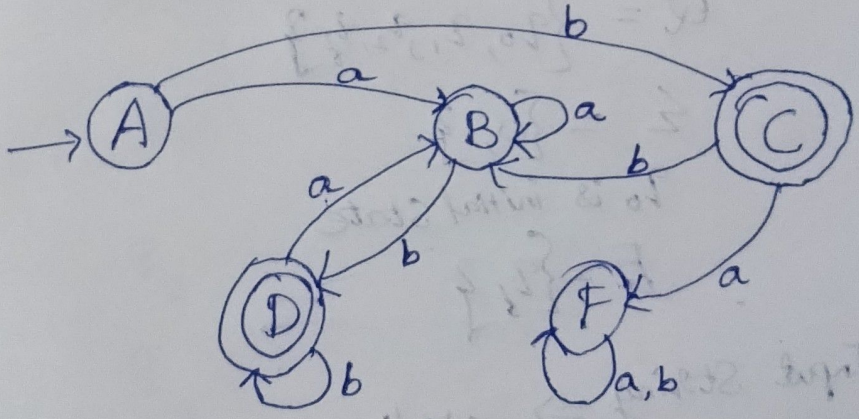
$A \xrightarrow{a} B$ $A \xrightarrow{b} C$

$B \xrightarrow{a} B$ $B \xrightarrow{b} D$

$C \xrightarrow{a} F$ $C \xrightarrow{b} B$

$D \xrightarrow{a} B$ $D \xrightarrow{b} D$

$F \xrightarrow{a,b} F$



(3) Design a Non-deterministic Finite Automata (NFA) that takes binary string as input and accepts the string only if last two digits are same (00 or 11). Write formal definition of NFA. Also show that string 01011 is accepted by NFA.

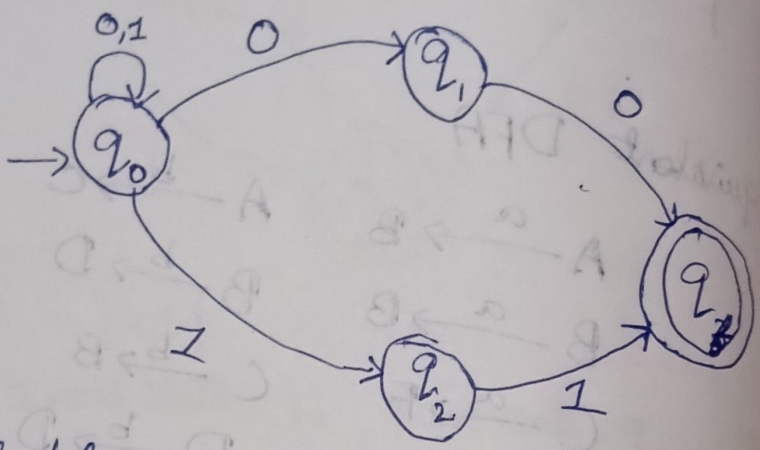
States :

q_0 - Initial State ; reads any num
of binary digits.

q_1 - NFA guesses 0 is second last dig

q_2 - NFA guesses 1 is second las

q_f - final State



NFA is defined as

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_f\}$$

$$\Sigma = \{0, 1\}$$

q_0 is initial state

$$F = \{q_f\}$$

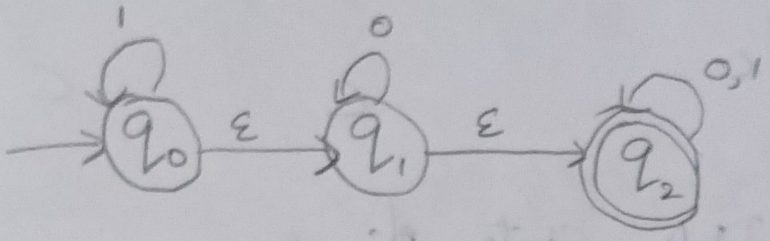
Input String => 01011

$$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$$

Final two digits are 11 & computation ends with q_f.

Therefore, 01011 is accepted by NFA

A NFA with ε- transitions is given below. Find ε-closure for each State and Design an equivalent DFA which accepts the same language. Draw transition table & mark initial and final States.



Given ε-NFA,

Initial State : q_0

Final State : q_2

q_0 has loop on input 1

$q_0 \xrightarrow{\varepsilon} q_2$

q_1 has loop on input 0

$q_1 \xrightarrow{\varepsilon} q_2$

q_2 has loop on inputs 0, 1.

State	ε-closure
q_0	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$
q_2	$\{q_2\}$

Equivalent DFA,

$A = \{q_0, q_1, q_2\}$ $C = \{q_2\}$

$B = \{q_1, q_2\}$

DFA State	Input 0	Input 1
→ * A	B	A
* B	B	C
* C	C	C

DFA State diagram

